

Probability and Mathematical Statistics

Theory, Methods, and Proofs for Data Science Admissions

A self-contained sequence of advanced lecture notes

The text develops probability from finite models and the Kolmogorov axioms through conditional expectation, concentration, limit theorems, estimation, testing, and regression. Every chapter is organized around definitions, reusable proof patterns, and worked examples at the level expected in a selective mathematical data-science program.

Preface

These notes are designed for a reader preparing for the mathematical component of a selective data-science program such as the Yandex School of Data Analysis. The core admissions material is combinatorics, elementary probability, random variables, expectations, and ingenious finite arguments. The later chapters provide the mathematical statistics needed to enter a serious data-science curriculum without a conceptual gap.

The presentation follows the style of an advanced MIT lecture sequence: definitions are precise, theorems are proved when the proof teaches a reusable method, and examples are chosen to expose structure rather than to rehearse arithmetic. This is an independent study text and is not an official MIT publication.

How to use the two volumes

Read one theory chapter, reproduce its principal definitions and proofs from memory, and then solve the corresponding chapter of the problem book. Problems marked *Warm-up* test definitions, *Core* problems test standard methods, *Advanced* problems require synthesis, and *Challenge* problems approximate the inventiveness of a strong admissions examination.

A complete solution specifies the probability model, identifies the random objects, justifies each independence or symmetry claim, and states the conditions of every theorem used. Numerical answers without a model are not complete arguments.

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Part I

Probability Foundations

CHAPTER 1

Counting Before Probability

Learning objectives

By the end of this chapter you should be able to choose the correct sample space, count it by products, combinations, recursion, or inclusion–exclusion, and recognize when a claimed symmetry does not produce equally likely outcomes.

Probability on a finite uniform space is a quotient of two counts. Consequently, a large fraction of “probability mistakes” are actually errors in the choice or enumeration of outcomes. We therefore begin with counting as a modeling language.

1.1 The sum and product principles

If a construction consists of r successive choices and the j th choice has n_j possibilities for every preceding history, then the number of constructions is

$$n_1 n_2 \cdots n_r.$$

If a set is partitioned into disjoint classes $A_1, \dots, A_r$, then

$$\left| \bigcup_{j=1}^r A_j \right| = \sum_{j=1}^r |A_j|.$$

The word *disjoint* is essential: without it, an object may be counted more than once.

How many binary strings of length n contain exactly k ones and no two ones adjacent? Write the string as k ones separated by at least one zero. Reserve $k - 1$ separating zeros. The remaining $n - (2k - 1)$ zeros may be placed in the $k + 1$ gaps before, between, and after the ones. Equivalently, shift the selected positions $1 \leq i_1 < \cdots < i_k \leq n$ to $j_r = i_r - (r - 1)$. Then

$$1 \leq j_1 < \cdots < j_k \leq n - k + 1,$$

so the answer is $\binom{n-k+1}{k}$.

1.2 Permutations, arrangements, and combinations

For $0 \leq k \leq n$,

$$n! = n(n-1) \cdots 1, \quad (n)_k = \frac{n!}{(n-k)!}, \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Here $(n)_k$ counts ordered samples without replacement, whereas $\binom{n}{k}$ counts unordered samples. If repetition is allowed, the number of ordered samples is n^k .

If n labeled objects are divided into labeled groups of sizes $n_1, \dots, n_r$, where $\sum n_i = n$, the number of divisions is

$$\binom{n}{n_1, \dots, n_r} = \frac{n!}{n_1! \cdots n_r!}.$$

Proof. Order all n objects, producing $n!$ lists. Permutations inside group i do not change the division, so every division occurs exactly $n_1! \cdots n_r!$ times. $\square$

1.2.1 Stars and bars

The number of nonnegative integer solutions of

$$x_1 + \cdots + x_r = n$$

is $\binom{n+r-1}{r-1}$. Represent n identical stars and insert $r-1$ bars. For positive solutions replace x_i by $y_i + 1$, obtaining $\binom{n-1}{r-1}$ when $n \geq r$.

1.3 Double counting and bijections

A combinatorial identity is often best proved by counting one set in two ways. For example,

$$\sum_k \binom{r}{k} \binom{s}{n-k} = \binom{r+s}{n}$$

because both sides count n -person committees drawn from groups of sizes r and s , with the left side conditioning on the number drawn from the first group.

For $1 \leq k \leq n$,

$$k \binom{n}{k} = n \binom{n-1}{k-1}.$$

Proof. Count pairs (C, c) consisting of a k -element committee C and a designated chair $c \in C$. Choose the committee then its chair, or choose the chair then the remaining $k-1$ members. $\square$

1.4 Inclusion–exclusion

For finite sets $A_1, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \cdots.$$

To see why, fix an element belonging to exactly m of the sets. Its total coefficient on the right is

$$\binom{m}{1} - \binom{m}{2} + \cdots + (-1)^{m+1} \binom{m}{m} = 1.$$

Let D_n be the number of permutations of n objects with no fixed point. Set $A_i = \{\pi : \pi(i) = i\}$. An intersection of k specified events has size $(n - k)!$. Hence

$$D_n = n! \sum_{k=0}^n \frac{(-1)^k}{k!}, \quad \frac{D_n}{n!} \rightarrow e^{-1}.$$

The proof reveals more than the formula: the number of fixed points in a large random permutation behaves approximately like a Poisson random variable with mean one.

1.5 Recurrences and complements

When a direct count is difficult, classify objects by their first move or last component. Let a_n denote binary strings of length n with no consecutive ones. A valid string ends in 0, following any valid length- $n - 1$ string, or ends in 10, following any valid length- $n - 2$ string. Thus $a_n = a_{n-1} + a_{n-2}$.

Counting the complement is particularly efficient for phrases such as “at least one.” Instead of enumerating all ways an event can happen, count the single structural way in which it fails.

The formula $\mathbb{P}(A) = |A|/|\Omega|$ is valid only when the elementary outcomes in Ω have equal probabilities. A random integer generated by first choosing its number of digits and then choosing its digits uniformly is not uniform over the integers produced.

Chapter check

You should now be able to explain the distinction between n^k , $(n)_k$, and $\binom{n}{k}$; derive stars and bars; use a bijection to prove an identity; and apply inclusion–exclusion without guessing signs.

CHAPTER 2

Probability Spaces and Events

Learning objectives

Translate an experiment into $(\Omega, \mathcal{F}, \mathbb{P})$, derive the basic calculus of events from the axioms, use continuity of probability, and distinguish finite uniform and geometric models.

2.1 The Kolmogorov model

Definition 2.1: Probability space

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ in which Ω is the sample space, $\mathcal{F}$ is a σ -algebra of subsets of Ω , and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfies

- (i) $\mathbb{P}(\Omega) = 1$;
- (ii) for pairwise disjoint $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} \mathbb{P}(A_n).$$

A σ -algebra contains Ω , is closed under complements, and is closed under countable unions. It specifies the questions to which the model assigns probabilities. On a finite or countable sample space we usually take $\mathcal{F} = 2^\Omega$; on an uncountable space this is generally impossible if we want a useful notion of length or volume.

2.2 Consequences of the axioms

For events A, B :

$$\begin{aligned} \mathbb{P}(\emptyset) &= 0, & \mathbb{P}(A^c) &= 1 - \mathbb{P}(A), \\ A \subseteq B &\implies \mathbb{P}(A) \leq \mathbb{P}(B), & \mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \end{aligned}$$

Moreover,

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

Proof. Since $\Omega = \Omega \sqcup \emptyset$, additivity gives $1 = 1 + \mathbb{P}(\emptyset)$. Since $\Omega = A \sqcup A^c$, the complement formula follows. If $A \subseteq B$, write $B = A \sqcup (B \setminus A)$. Finally, decompose $A \cup B = A \sqcup (B \setminus A)$ and use $B = (A \cap B) \sqcup (B \setminus A)$. The union bound follows by replacing A_i with the disjoint sets $A_i \setminus \bigcup_{j < i} A_j$. $\square$

2.3 Continuity of probability

If $A_n \uparrow A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$. If $A_n \downarrow A$, then $\mathbb{P}(A_n) \rightarrow \mathbb{P}(A)$.

Proof. For the increasing case define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$. The B_n are disjoint, $A = \bigcup B_n$, and $A_m = \bigcup_{n \leq m} B_n$. Countable additivity turns the assertion into convergence of partial sums. The decreasing case follows by applying the increasing case to complements. $\square$

Suppose independent trials have success probabilities p_n . The event that at least one success occurs after time m is

$$B_m = \bigcup_{n \geq m} \{\text{success at } n\}.$$

The events B_m decrease, and $\bigcap_m B_m$ is the event of infinitely many successes. Therefore

$$\mathbb{P}(\text{infinitely many successes}) = \lim_{m \rightarrow \infty} \mathbb{P}(B_m).$$

This reduction is the first step toward the Borel–Cantelli lemmas.

2.4 Finite uniform models

If Ω is finite and $\mathbb{P}(\{\omega\}) = 1/|\Omega|$ for every ω , then

$$\mathbb{P}(A) = \frac{|A|}{|\Omega|}.$$

The sample space should retain exactly the information that makes elementary outcomes equiprobable. Five-card poker hands are unordered 5-subsets, whereas five successive draws are ordered 5-tuples without repetition. Both models work, but numerator and denominator must use the same convention.

Assume n birthdays are independent and uniform among 365 days. The probability of no collision is

$$\frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n},$$

so the collision probability is one minus this expression. Counting the complement avoids a complicated classification by collision pattern.

2.5 Geometric probability

If a random point is uniform in a region $D \subset \mathbb{R}^d$ of finite positive volume, then

$$\mathbb{P}(X \in A) = \frac{\text{vol}(A \cap D)}{\text{vol}(D)}.$$

Uniformity is coordinate dependent under nonlinear maps. A uniform radius $R \in [0, 1]$ does not generate a uniform point in the unit disk, because equal radial increments correspond to annuli of unequal area. A uniform point in the disk has $\mathbb{P}(R \leq r) = r^2$ and density $2r$.

Before calculating, answer three questions: What is an elementary outcome? Why are the outcomes equally likely, or what measure replaces counting? Does the event depend only on information retained by the sample space?

Chapter check

Derive all four identities in Proposition 2.2 from the axioms. Explain why continuity from above requires $\mathbb{P}(A_1) < \infty$ for general measures but needs no extra assumption for probabilities.

CHAPTER 3

Conditional Probability and Bayes' Rule

Learning objectives

Update a probability model after information arrives, use multiplication and total probability, derive Bayes' rule, and solve multistage problems by conditioning on a well-chosen latent variable.

3.1 Conditioning as renormalization

Definition 3.1: Conditional probability

For events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed B , the map $A \mapsto \mathbb{P}(A | B)$ is a probability measure. Informally, information that B occurred removes B^c from consideration and rescales the remaining probabilities to sum to one.

If all conditioning events have positive probability, then

$$\mathbb{P}(A_1 \cap \dots \cap A_n) = \mathbb{P}(A_1) \prod_{j=2}^n \mathbb{P}(A_j | A_1 \cap \dots \cap A_{j-1}).$$

Proof. Apply $\mathbb{P}(C \cap D) = \mathbb{P}(C)\mathbb{P}(D | C)$ repeatedly. All intermediate factors cancel if the definitions are expanded. $\square$

3.2 Partitions and total probability

Let $H_1, \dots, H_m$ be a partition of Ω with positive probabilities. Since A is the disjoint union of $A \cap H_i$,

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A | H_i)\mathbb{P}(H_i). \quad (3.1)$$

The H_i are often competing sources, types, regimes, or values of a first-stage random choice.

Under the preceding assumptions, for $\mathbb{P}(A) > 0$,

$$\mathbb{P}(H_j | A) = \frac{\mathbb{P}(A | H_j)\mathbb{P}(H_j)}{\sum_i \mathbb{P}(A | H_i)\mathbb{P}(H_i)}.$$

Proof. The numerator of the definition is $\mathbb{P}(A \cap H_j) = \mathbb{P}(A | H_j)\mathbb{P}(H_j)$; substitute (3.1) in the denominator. $\square$

A disease has prevalence 1%. A test has sensitivity 95% and specificity 90%. For a positive result +,

$$\mathbb{P}(D | +) = \frac{0.95(0.01)}{0.95(0.01) + 0.10(0.99)} \approx 0.0876.$$

The posterior is far below the sensitivity because false positives arise from a much larger healthy population. Base rates cannot be discarded.

3.3 Odds form and sequential evidence

For two hypotheses H_1, H_0 , Bayes' rule can be written as

$$\frac{\mathbb{P}(H_1 | A)}{\mathbb{P}(H_0 | A)} = \frac{\mathbb{P}(H_1)}{\mathbb{P}(H_0)} \frac{\mathbb{P}(A | H_1)}{\mathbb{P}(A | H_0)}.$$

Posterior odds equal prior odds times the likelihood ratio. For conditionally independent observations, likelihood ratios multiply, making log-odds additive.

3.4 Condition on what makes the problem simple

Conditioning is not merely a formula; it is a decomposition strategy. A good conditioning variable freezes the source of heterogeneity.

A fair die is rolled; then a fair coin is tossed as many times as the die shows. Let H be the number of heads and N the die value. Conditional on $N = n$, $H \sim \text{Bin}(n, 1/2)$, so

$$\mathbb{P}(H = 0) = \frac{1}{6} \sum_{n=1}^6 2^{-n}, \quad \mathbb{E}H = \mathbb{E}(\mathbb{E}[H | N]) = \mathbb{E}(N/2) = \frac{7}{4}.$$

The first identity is total probability; the second anticipates conditional expectation.

3.5 Information protocols matter

Suppose one of prisoners A, B, C will be released uniformly. Prisoner A asks a guard to name another prisoner who will not be released; the guard says B . The posterior probability that A is released cannot be computed until we know how the guard chooses between B and C when A is released. Conditional probability is a property of the full experiment, including the information mechanism.

Common pitfall. Confusing $\mathbb{P}(A | B)$ with $\mathbb{P}(B | A)$. The two quantities use different denominators. Bayes' rule connects them; symmetry of notation does not.

Name the hypotheses; write their prior probabilities; compute the likelihood of the observation under each hypothesis; normalize. This structure is usually clearer than drawing a large tree.

Chapter check

Explain total probability as a weighted average. Derive Bayes' rule rather than memorizing it. Give an example in which the posterior depends on the protocol used to reveal information.

CHAPTER 4

Independence

Learning objectives

Distinguish disjointness from independence, verify mutual rather than merely pairwise independence, understand conditional independence, and model repeated trials correctly.

4.1 Events and information

Definition 4.1: Independent events

Events A and B are independent if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

If $\mathbb{P}(B) > 0$, this is equivalent to $\mathbb{P}(A | B) = \mathbb{P}(A)$.

Disjoint non-null events cannot be independent: disjointness says they cannot occur together, whereas independence says learning one occurred gives no information about the other.

If A and B are independent, then each pair chosen from A, A^c and B, B^c is independent.

Proof. For instance,

$$\mathbb{P}(A^c \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = \mathbb{P}(B)(1 - \mathbb{P}(A)) = \mathbb{P}(A^c)\mathbb{P}(B).$$

The other cases follow similarly. □

4.2 Families of events

Definition 4.3: Mutual independence

Events $A_1, \dots, A_n$ are mutually independent if for every nonempty $I \subseteq \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

Checking only pairs is insufficient.

Let X, Y be independent fair bits and set $Z = X \oplus Y$. Then X, Y, Z are pairwise independent:

fixing any one bit leaves each other bit fair. Yet Z is determined by (X, Y) , and

$$\mathbb{P}(X = 0, Y = 0, Z = 0) = \frac{1}{4} \neq \frac{1}{8}.$$

Pairwise independence controls second-order interactions but not higher-order constraints.

4.3 Independent random variables

Random variables $X_1, \dots, X_n$ are independent if the events $\{X_i \in B_i\}$ are mutually independent for all Borel sets B_i . In the discrete case this is equivalent to factorization of the joint mass function; for variables with a joint density it is equivalent to

$$f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

almost everywhere.

If random vectors X and Y are independent, then $g(X)$ and $h(Y)$ are independent for all measurable functions g, h .

This fact legitimizes grouping independent experiments into derived statistics. Two functions of the *same* variable are generally dependent.

4.4 Bernoulli trials

In n independent trials with success probability p , the number of successes

$$S_n = I_1 + \dots + I_n$$

has distribution

$$\mathbb{P}(S_n = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The binomial coefficient chooses the successful positions; independence gives the probability of each fixed pattern.

For $0 \leq k < n$,

$$\frac{\mathbb{P}(S_n = k+1)}{\mathbb{P}(S_n = k)} = \frac{n-k}{k+1} \frac{p}{1-p}.$$

The probabilities increase while $k+1 \leq (n+1)p$ and decrease afterward. Thus the mode is $\lfloor (n+1)p \rfloor$, except that if $(n+1)p$ is an integer, two adjacent values are modes.

4.5 Conditional independence and mixtures

Variables X, Y are conditionally independent given Z if their conditional joint law factorizes for each relevant value of Z . Conditional independence does not imply unconditional independence. If Z selects a population and X, Y both respond to Z , then mixing the populations creates association even when X and Y are independent within each population. This is one mechanism behind Simpson's paradox.

Conversely, unconditional independence can disappear after conditioning. Two independent fair coin tosses become dependent given that their total number of heads is one.

Independence belongs to a probability model. Shared random parameters, sampling without replacement, or conditioning on a common outcome can couple physically separate objects.

Chapter check

Construct three pairwise independent events that are not mutually independent. Prove complement preservation. Explain why sampling with replacement is independent while sampling without replacement is not.

Part II

Random Variables and Their Calculus

CHAPTER 5

Random Variables and Distribution Functions

Learning objectives

Treat a random variable as a measurable function, move fluently among mass functions, densities, and cumulative distribution functions, compute quantiles, and derive distributions of simple transformations.

5.1 Random variables are functions

Definition 5.1: Random variable

A real-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is a measurable function $X : \Omega \rightarrow \mathbb{R}$; that is, $\{X \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

The randomness lies in the outcome ω ; once ω is fixed, $X(\omega)$ is an ordinary number. The distribution of X is the probability measure

$$\mu_X(B) = \mathbb{P}(X \in B),$$

also called the pushforward of $\mathbb{P}$ by X . Different experiments may produce random variables with the same distribution.

5.2 The cumulative distribution function

Definition 5.2: Distribution function

The cumulative distribution function (CDF) of X is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

Conversely, every function with these properties is the CDF of a probability distribution on $\mathbb{R}$.

Right-continuity follows from continuity of probability applied to $\{X \leq x + 1/n\} \downarrow \{X \leq x\}$. The jump at x is

$$F_X(x) - F_X(x^-) = \mathbb{P}(X = x).$$

5.3 Discrete and continuous laws

If X takes countably many values x_k , its probability mass function is $p_X(x_k) = \mathbb{P}(X = x_k)$ and $\sum_k p_X(x_k) = 1$. If there exists $f_X \geq 0$ such that

$$F_X(x) = \int_{-\infty}^x f_X(t) dt,$$

then f_X is a density. Densities are not unique at individual points, and $f_X(x)$ is not $\mathbb{P}(X = x)$; every point has probability zero for a continuous law.

Not every distribution is purely discrete or purely continuous. A waiting-time model may have an atom at zero and a density on $(0, \infty)$.

Suppose $f(x) = cx(1-x)$ on $[0, 1]$ and zero elsewhere. Normalization gives

$$1 = c \int_0^1 x(1-x) dx = c \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{c}{6},$$

so $c = 6$. Then

$$F(x) = 3x^2 - 2x^3, \text{ for } 0 \leq x \leq 1.$$

5.4 Quantiles

For $0 < \alpha < 1$, a standard definition is

$$q_\alpha = \inf\{x : F_X(x) \geq \alpha\}.$$

The median is a $1/2$ -quantile. In a discrete distribution, a quantile need not satisfy $F(q_\alpha) = \alpha$ exactly. Quantiles transform naturally under increasing functions: $q_\alpha(g(X)) = g(q_\alpha(X))$ when g is strictly increasing and regular enough.

5.5 Transformations via the CDF

To find the law of $Y = g(X)$, begin with

$$F_Y(y) = \mathbb{P}(g(X) \leq y).$$

If g is strictly increasing and differentiable,

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

For a non-injective map, sum the contributions of all inverse branches.

Let $X \sim \text{Unif}[-1, 1]$ and $Y = X^2$. For $0 \leq y \leq 1$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y}, \quad f_Y(y) = \frac{1}{2\sqrt{y}}.$$

Both inverse branches $\pm\sqrt{y}$ contribute; the CDF method handles them automatically.

Common pitfall. Writing $f_Y(y) = f_X(g^{-1}(y))$ and forgetting the derivative. A density is probability per unit length, so it must change when the scale changes.

Chapter check

State the four defining properties of a CDF. Explain atoms as jumps. Find the distribution of $|X|$ in terms of F_X and simplify when X has a symmetric density.

CHAPTER 6

Expectation and Variance

Learning objectives

Compute expectations without first computing full distributions, use indicators and tail sums, manipulate variance and covariance correctly, and recognize when an expectation fails to exist.

6.1 Expectation as integration

For a discrete random variable,

$$\mathbb{E}X = \sum_x x\mathbb{P}(X = x),$$

provided the positive and negative parts do not both have infinite expectation. For a variable with density,

$$\mathbb{E}X = \int_{-\infty}^{\infty} xf_X(x) dx.$$

More generally, expectation is integration with respect to the probability measure: $\mathbb{E}X = \int X d\mathbb{P}$.

For a measurable g , whenever the expectation exists,

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int g(x)f_X(x) dx.$$

One need not first derive the distribution of $g(X)$.

6.2 Linearity and indicators

For integrable random variables $X_1, \dots, X_n$ and constants a_i ,

$$\mathbb{E} \left[\sum_{i=1}^n a_i X_i \right] = \sum_{i=1}^n a_i \mathbb{E}X_i.$$

Independence is not required.

For an event A , let $\mathbf{1}_A$ be one on A and zero otherwise. Then $\mathbb{E}\mathbf{1}_A = \mathbb{P}(A)$. Counts can therefore be written as sums of indicators.

For a uniform random permutation π of $1, \dots, n$, let

$$I_{ij} = \mathbf{1}_{\{\pi(i) > \pi(j)\}}, \quad i < j.$$

The number of inversions is $N = \sum_{i < j} I_{ij}$. By symmetry $\mathbb{E}I_{ij} = 1/2$, hence

$$\mathbb{E}N = \binom{n}{2} \frac{1}{2} = \frac{n(n-1)}{4}.$$

The indicators are dependent, but linearity does not care.

6.3 Tail-sum formulas

If X is nonnegative, then

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t) dt.$$

If X is nonnegative and integer-valued,

$$\mathbb{E}X = \sum_{k=1}^\infty \mathbb{P}(X \geq k).$$

Proof. Pointwise, $X(\omega) = \int_0^\infty \mathbf{1}_{\{X(\omega) > t\}} dt$. Interchange the nonnegative integrals. In the integer case, $X = \sum_{k \geq 1} \mathbf{1}_{\{X \geq k\}}$. $\square$

6.4 Variance and covariance

Definition 6.4: Variance and covariance

For square-integrable variables,

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2, \\ \text{Cov}(X, Y) &= \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y. \end{aligned}$$

The central algebraic identity is

$$\text{Var}\left(\sum_i X_i\right) = \sum_i \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Independent square-integrable variables have zero covariance, but zero covariance does not imply independence.

If $\text{Var}(X), \text{Var}(Y) > 0$, then

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} \in [-1, 1].$$

Equality in absolute value holds exactly when $Y = aX + b$ almost surely for constants $a \neq 0, b$.

Proof. Apply Cauchy–Schwarz to $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$. Equality in Cauchy–Schwarz is linear dependence in L^2 . $\square$

6.5 Expectation as an existence tool

If a nonnegative integer-valued random variable N satisfies $\mathbb{E}N < 1$, then some outcome has $N = 0$. This elementary observation is the core of the probabilistic method.

Place every vertex of a graph independently into one of two parts with probability $1/2$. Each edge crosses the cut with probability $1/2$. Thus the expected cut size is $|E|/2$, so at least one cut has size at least $|E|/2$.

Common pitfall. The formula $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ needs zero covariance. Linearity of expectation needs no independence; additivity of variance does.

Chapter check

Prove the integer tail-sum formula. Compute the expected number of fixed points in a random permutation. Give dependent, uncorrelated random variables.

CHAPTER 7

Canonical Discrete Distributions

Learning objectives

Recognize Bernoulli, binomial, geometric, negative-binomial, hypergeometric, and Poisson models; derive their moments; and move between sampling descriptions and formulas.

7.1 Bernoulli and binomial laws

For $X \sim \text{Bern}(p)$,

$$\mathbb{P}(X = 1) = p, \quad \mathbb{E}X = p, \quad \text{Var}(X) = p(1 - p).$$

If $X_1, \dots, X_n$ are independent $\text{Bern}(p)$ variables, their sum is $\text{Bin}(n, p)$ with

$$\mathbb{P}(S = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E}S = np, \quad \text{Var}(S) = np(1 - p).$$

7.2 Sampling without replacement

If n objects are drawn without replacement from a population of N containing K marked objects, then

$$X \sim \text{Hypergeom}(N, K, n), \quad \mathbb{P}(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}.$$

Each draw has marginal success probability K/N , so indicator linearity gives $\mathbb{E}X = nK/N$. The draws are negatively dependent, and

$$\text{Var}(X) = n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N - n}{N - 1}.$$

The final factor is the finite-population correction.

7.3 Geometric and negative-binomial laws

Let T be the trial number of the first success in independent Bernoulli trials. Then

$$\mathbb{P}(T = k) = (1 - p)^{k-1} p, \quad \mathbb{P}(T > m) = (1 - p)^m, \quad \mathbb{E}T = \frac{1}{p}.$$

For $m, n \geq 0$,

$$\mathbb{P}(T > m + n \mid T > m) = \mathbb{P}(T > n).$$

The geometric distribution is the only nondegenerate distribution on $\{1, 2, \dots\}$ with this property.

The number of trials required for the r th success has negative-binomial mass

$$\mathbb{P}(T_r = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k \geq r.$$

The last trial must be a success; choose the other $r-1$ successes among the first $k-1$ trials.

7.4 Poisson counts

For $X \sim \text{Pois}(\lambda)$,

$$\mathbb{P}(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad \mathbb{E}X = \text{Var}(X) = \lambda.$$

The recurrence $\mathbb{P}(X = k+1) = \lambda \mathbb{P}(X = k)/(k+1)$ makes both computation and mode identification simple.

If $X \sim \text{Pois}(\lambda)$ and $Y \sim \text{Pois}(\mu)$ are independent, then $X + Y \sim \text{Pois}(\lambda + \mu)$.

Proof. For $n \geq 0$, convolution and the binomial theorem give

$$\begin{aligned} \mathbb{P}(X + Y = n) &= e^{-(\lambda+\mu)} \sum_{k=0}^n \frac{\lambda^k \mu^{n-k}}{k!(n-k)!} \\ &= e^{-(\lambda+\mu)} \frac{(\lambda + \mu)^n}{n!}. \end{aligned}$$

□

Let $N \sim \text{Pois}(\lambda)$. Conditional on N , retain each item independently with probability p , and call the retained count X . Then

$$\mathbb{P}(X = k) = \sum_{n \geq k} \binom{n}{k} p^k (1-p)^{n-k} e^{-\lambda} \frac{\lambda^n}{n!} = e^{-\lambda p} \frac{(\lambda p)^k}{k!}.$$

Thus $X \sim \text{Pois}(\lambda p)$. The discarded count is $\text{Pois}(\lambda(1-p))$, and the two counts are independent.

7.5 Selecting a model

Description	Model
One binary trial	Bernoulli
Fixed number of independent binary trials	Binomial
Fixed-size sample without replacement	Hypergeometric
Trials until first or r th success	Geometric / negative binomial
Rare independent events in time or space	Poisson

“With replacement” suggests independence and a binomial law; “without replacement” suggests a hypergeometric law. “Until” signals a waiting-time law. “Per unit time” suggests Poisson only if independent increments and a constant rate are defensible.

Chapter check

Derive each mass function from the experiment rather than memorizing it. Explain why the hypergeometric variance is smaller than the binomial variance with the same marginal success probability.

CHAPTER 8

Canonical Continuous Distributions

Learning objectives

Recognize uniform, exponential, gamma, beta, and normal models; compute probabilities and moments; use memorylessness and standardization; and understand how these families arise structurally.

8.1 Uniform distributions

For $X \sim \text{Unif}[a, b]$,

$$f_X(x) = \frac{1}{b-a} \mathbf{1}_{[a,b]}(x), \quad \mathbb{E}X = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Uniform means probability is proportional to length in the chosen coordinate. If $U \sim \text{Unif}[0, 1]$ and F is a continuous strictly increasing CDF, then $F^{-1}(U)$ has CDF F . This inverse-transform construction is fundamental in simulation.

8.2 Exponential waiting times

For $X \sim \text{Exp}(\lambda)$,

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{x \geq 0}, \quad \mathbb{P}(X > t) = e^{-\lambda t},$$

and

$$\mathbb{E}X = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$$

For $s, t \geq 0$,

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

The exponential family is the only nondegenerate continuous family on $[0, \infty)$ with this property, under mild regularity.

If $X_i \sim \text{Exp}(\lambda_i)$ are independent, then

$$\min_i X_i \sim \text{Exp}\left(\sum_i \lambda_i\right), \quad \mathbb{P}(X_j = \min_i X_i) = \frac{\lambda_j}{\sum_i \lambda_i}.$$

These identities encode competing exponential clocks.

8.3 Gamma and beta families

For shape $\alpha > 0$ and rate $\lambda > 0$,

$$f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x > 0.$$

Then $\mathbb{E}X = \alpha/\lambda$ and $\text{Var}(X) = \alpha/\lambda^2$. A sum of independent exponential variables with common rate λ is gamma; for integer shape it is the waiting time to the α th event in a Poisson process.

For $X \sim \text{Beta}(\alpha, \beta)$,

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < x < 1,$$

with $\mathbb{E}X = \alpha/(\alpha + \beta)$. The beta family is natural for random proportions and for Bayesian models of Bernoulli probabilities.

8.4 The normal distribution

For $X \sim \mathcal{N}(\mu, \sigma^2)$,

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right].$$

The standardized variable $Z = (X - \mu)/\sigma$ is $\mathcal{N}(0, 1)$. Symmetry gives $\Phi(-z) = 1 - \Phi(z)$.

If X_i are independent normal variables and a_i, b are constants, then

$$b + \sum_i a_i X_i \sim \mathcal{N}\left(b + \sum_i a_i \mu_i, \sum_i a_i^2 \sigma_i^2\right).$$

One proof uses moment-generating or characteristic functions; a geometric proof uses rotational symmetry of the standard multivariate normal.

If a measurement is $\mathcal{N}(100, 15^2)$, then

$$\mathbb{P}(85 \leq X \leq 130) = \mathbb{P}(-1 \leq Z \leq 2) = \Phi(2) - \Phi(-1).$$

Standardization is a change of units, not an approximation.

8.5 Why the normal law is central

The normal density is the maximum-entropy density among all real distributions with specified mean and variance. It is stable under independent addition and emerges as the universal limit of standardized sums under broad conditions. These are structural reasons, not merely empirical convenience.

Common pitfall. The parameter in $\mathcal{N}(\mu, \sigma^2)$ is the variance, not the standard deviation. For the exponential law, verify whether a source uses rate λ or mean/scale $1/\lambda$.

Chapter check

Derive the mean of $\text{Unif}[a, b]$ by symmetry and its variance by integration. Prove the minimum-of-exponentials formula from survival functions. Explain standardization in words.

CHAPTER 9

Joint Distributions and Random Vectors

Learning objectives

Recover marginals and conditionals from a joint law, test independence by factorization and support, use covariance matrices, and interpret correlation without confusing it with causation or general dependence.

9.1 Joint, marginal, and conditional laws

For a random vector (X, Y) the joint CDF is

$$F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y).$$

In the discrete case a joint mass function $p(x, y)$ satisfies $\sum_{x,y} p(x, y) = 1$, and marginals are obtained by summation:

$$p_X(x) = \sum_y p(x, y), \quad p_Y(y) = \sum_x p(x, y).$$

For a joint density,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

When $f_Y(y) > 0$, the conditional density is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

Suppose (X, Y) is uniform on $D = \{(x, y) : 0 < x < y < 1\}$. Since $\text{area}(D) = 1/2$, the joint density is 2 on D . Thus

$$f_X(x) = \int_x^1 2 dy = 2(1 - x), \quad f_Y(y) = \int_0^y 2 dx = 2y.$$

The triangular support already proves dependence: a product support would be rectangular up to null sets.

9.2 Independence tests

For discrete variables, independence is equivalent to $p(x, y) = p_X(x)p_Y(y)$ for all x, y . For densities the equality need hold only almost everywhere. Factorization of an unnormalized expression is not enough if the support does not factor.

If X and Y are independent and $g(X), h(Y)$ are integrable, then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)].$$

The converse, if it holds for every bounded measurable g, h , characterizes independence. Checking only $g(x) = x$ and $h(y) = y$ checks zero covariance, not independence.

9.3 Covariance matrices

For $X = (X_1, \dots, X_d)^\top$ with finite second moments, define

$$\mu = \mathbb{E}X, \text{ and } \Sigma = \mathbb{E}[(X - \mu)(X - \mu)^\top].$$

Then $\Sigma_{ij} = \text{Cov}(X_i, X_j)$ and, for every $a \in \mathbb{R}^d$,

$$\text{Var}(a^\top X) = a^\top \Sigma a \geq 0.$$

Thus every covariance matrix is symmetric positive semidefinite. If $Y = AX + b$, then

$$\mathbb{E}Y = A\mu + b, \text{ and } \text{Cov}(Y) = A\Sigma A^\top.$$

Suppose X, Y estimate the same quantity, are unbiased, and have variances σ_X^2, σ_Y^2 and covariance c . An unbiased combination is $T_a = aX + (1-a)Y$. Its variance is a quadratic in a :

$$a^2\sigma_X^2 + (1-a)^2\sigma_Y^2 + 2a(1-a)c.$$

Differentiation yields the minimum-variance weight

$$a^* = \frac{\sigma_Y^2 - c}{\sigma_X^2 + \sigma_Y^2 - 2c},$$

provided the denominator is positive.

9.4 Correlation: power and limits

Correlation is invariant under positive affine changes of scale and measures linear association. It can be zero under strong nonlinear dependence: if X is symmetric about zero and $Y = X^2$, then $\text{Cov}(X, Y) = \mathbb{E}X^3 = 0$ although Y is determined by X .

Observed correlation also does not identify a causal relation. Confounding, selection, feedback, and aggregation can generate or reverse associations. Probability supplies the language for association; causal conclusions require additional assumptions or designs.

Before integrating, sketch or state the support. It determines the limits, often reveals dependence instantly, and prevents the most common error in multivariate calculations.

Chapter check

Compute marginals from a joint density on a nonrectangular support. Prove Σ is positive semidefinite. Give two reasons why zero correlation is weaker than independence.

CHAPTER 10

Transformations, Convolution, and Order Statistics

Learning objectives

Find laws of sums and extrema, apply the multivariate Jacobian formula with the correct support, and derive distributions of order statistics.

10.1 Convolution

If X, Y are independent discrete variables,

$$\mathbb{P}(X + Y = z) = \sum_x \mathbb{P}(X = x) \mathbb{P}(Y = z - x).$$

For independent variables with densities,

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z - x) dx.$$

The integration interval is where both density factors are nonzero.

For independent $X, Y \sim \text{Unif}[0, 1]$, the integrand is one when $0 \leq x \leq 1$ and $0 \leq z - x \leq 1$. The length of this intersection is

$$f_{X+Y}(z) = \begin{cases} z, & 0 \leq z \leq 1, \\ 2 - z, & 1 < z \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

The triangular density is geometry disguised as convolution.

10.2 Change of variables in several dimensions

Let $(U, V) = g(X, Y)$ be a differentiable one-to-one transformation with inverse $(x(u, v), y(u, v))$. Then

$$f_{U,V}(u, v) = f_{X,Y}(x(u, v), y(u, v)) \left| \det \frac{\partial(x, y)}{\partial(u, v)} \right|$$

on the transformed support.

Let X, Y be independent $\text{Exp}(1)$ variables and define

$$U = X + Y, \text{quad} V = \frac{X}{X + Y}.$$

The inverse is $X = UV$, $Y = U(1 - V)$, with Jacobian magnitude U . Hence

$$f_{U,V}(u, v) = e^{-u}u, \quad u > 0, \quad 0 < v < 1.$$

This factors into a $\text{Gamma}(2, 1)$ density for U and a $\text{Unif}[0, 1]$ density for V , so U and V are independent.

10.3 Minima and maxima

For independent identically distributed variables with CDF F ,

$$\begin{aligned} \mathbb{P}(X_{(n)} \leq x) &= F(x)^n, \\ \mathbb{P}(X_{(1)} > x) &= [1 - F(x)]^n. \end{aligned}$$

Here $X_{(1)} = \min_i X_i$ and $X_{(n)} = \max_i X_i$. Complementary events make these formulas nearly automatic.

If $X_i \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]$, then

$$f_{X_{(n)}}(x) = nx^{n-1}, \quad \mathbb{E}X_{(n)} = \frac{n}{n+1}.$$

Similarly $\mathbb{E}X_{(1)} = 1/(n+1)$. The expected gaps from the endpoints are equal by symmetry.

10.4 The k th order statistic

For a continuous distribution, $X_{(k)} \leq x$ exactly when at least k observations are at most x . Thus

$$\mathbb{P}(X_{(k)} \leq x) = \sum_{j=k}^n \binom{n}{j} F(x)^j [1 - F(x)]^{n-j}.$$

Differentiation gives

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1} [1 - F(x)]^{n-k} f(x).$$

Equivalently, $F(X_{(k)}) \sim \text{Beta}(k, n+1-k)$.

10.5 Random sums

If $S_N = X_1 + \dots + X_N$, where N is independent of iid X_i and all expectations exist, then conditioning on N yields

$$\mathbb{E}S_N = \mathbb{E}N \mathbb{E}X_1.$$

With finite variances,

$$\text{Var}(S_N) = \mathbb{E}N \text{Var}(X_1) + \text{Var}(N)(\mathbb{E}X_1)^2.$$

These are the mean and variance forms of Wald's decomposition.

Common pitfall. The Jacobian formula is incomplete without the transformed support. In many exam solutions, the determinant is correct and the region is wrong.

Chapter check

Derive convolution from conditioning. Obtain the maximum CDF in one line. Explain why $F(X)$ is uniform when F is continuous and strictly increasing.

CHAPTER 11

Conditional Expectation

Learning objectives

Use conditional expectation as a random variable, apply tower and pull-out properties, decompose variance, solve recurrences by first-step analysis, and interpret conditioning as prediction.

11.1 From conditional probabilities to random variables

For discrete Y , define

$$\mathbb{E}[X \mid Y = y] = \sum_x x \mathbb{P}(X = x \mid Y = y).$$

Then $\mathbb{E}[X \mid Y]$ is the random variable obtained by evaluating this function at the observed Y . In full generality, $\mathbb{E}[X \mid \mathcal{G}]$ is a $\mathcal{G}$ -measurable random variable satisfying

$$\mathbb{E}[X \mathbf{1}_G] = \mathbb{E}[\mathbb{E}(X \mid \mathcal{G}) \mathbf{1}_G] \quad \text{for every } G \in \mathcal{G}.$$

It is unique almost surely.

11.2 The working rules

For integrable variables:

- (i) linearity: $\mathbb{E}[aX + bY \mid \mathcal{G}] = a\mathbb{E}[X \mid \mathcal{G}] + b\mathbb{E}[Y \mid \mathcal{G}]$;
- (ii) pull-out: if Z is $\mathcal{G}$ -measurable, $\mathbb{E}[ZX \mid \mathcal{G}] = Z\mathbb{E}[X \mid \mathcal{G}]$;
- (iii) tower: if $\mathcal{H} \subseteq \mathcal{G}$, then $\mathbb{E}[\mathbb{E}(X \mid \mathcal{G}) \mid \mathcal{H}] = \mathbb{E}[X \mid \mathcal{H}]$;
- (iv) total expectation: $\mathbb{E}\mathbb{E}[X \mid \mathcal{G}] = \mathbb{E}X$.

If $N \sim \text{Pois}(\lambda)$ and $X \mid N \sim \text{Bin}(N, p)$, then

$$\mathbb{E}X = \mathbb{E}[Np] = \lambda p.$$

Also

$$\text{Var}(X) = \mathbb{E}[Np(1-p)] + \text{Var}(Np) = \lambda p(1-p) + \lambda p^2 = \lambda p.$$

Mean and variance match the $\text{Pois}(\lambda p)$ law.

11.3 The law of total variance

For $X \in L^2$,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X | \mathcal{G})] + \text{Var}(\mathbb{E}[X | \mathcal{G}]).$$

Proof. Write $X - \mathbb{E}X = (X - \mathbb{E}[X | \mathcal{G}]) + (\mathbb{E}[X | \mathcal{G}] - \mathbb{E}X)$. The cross term has expectation zero by the pull-out and tower properties. Expand the square. $\square$

The first term is average residual noise after observing $\mathcal{G}$; the second is the variability explained by that information. In particular,

$$\text{Var}(\mathbb{E}[X | \mathcal{G}]) \leq \text{Var}(X).$$

11.4 Best prediction in squared error

Among all square-integrable $\mathcal{G}$ -measurable random variables Z , the conditional expectation $m = \mathbb{E}[X | \mathcal{G}]$ minimizes $\mathbb{E}[(X - Z)^2]$.

Proof. The error decomposes orthogonally:

$$\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - m)^2] + \mathbb{E}[(m - Z)^2],$$

because $X - m$ is orthogonal to every square-integrable $\mathcal{G}$ -measurable variable. $\square$

11.5 First-step analysis

Let T be a waiting time in a process. Conditioning on the first transition often gives

$$t_i = 1 + \sum_j p_{ij} t_j$$

for expected remaining times t_i , with boundary conditions $t_i = 0$ at terminal states.

Let e_0 be the expected additional tosses with no useful suffix and e_1 after a final head. Then

$$e_0 = 1 + \frac{1}{2}e_1 + \frac{1}{2}e_0, \text{ and } e_1 = 1 + \frac{1}{2} \cdot 0 + \frac{1}{2}e_0.$$

Solving gives $e_0 = 6$. States encode exactly the information from the past needed for the future.

For pattern waiting problems, the state is the longest suffix of the observed sequence that is also a prefix of the target pattern. Keeping the entire history is unnecessary; keeping too little makes the recursion wrong.

Chapter check

State the tower property in words. Derive total variance. Set up, but need not solve, the state equations for waiting until the pattern HTH appears.

Part III

Bounds, Asymptotics, and Stochastic Models

CHAPTER 12

Probability Inequalities and the Probabilistic Method

Learning objectives

Select Markov, Chebyshev, Jensen, Cauchy–Schwarz, or a union bound; derive exponential bounds by the Chernoff method; and turn expectation estimates into existence proofs.

12.1 Markov and Chebyshev

If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X}{a}.$$

Proof. Pointwise $X \geq a\mathbf{1}_{\{X \geq a\}}$. Take expectations. □

Applying Markov to $(X - \mathbb{E}X)^2$ gives

If X has finite variance, then

$$\mathbb{P}(|X - \mathbb{E}X| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

These bounds are distribution-free and therefore sometimes loose. Their importance is that they require very little information.

12.2 Union bounds and Bonferroni inequalities

For any events,

$$\mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

The first two inclusion–exclusion truncations give

$$\sum_i \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) \leq \mathbb{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathbb{P}(A_i).$$

The union bound needs no independence and is often effective for rare failures.

12.3 Convexity inequalities

If φ is convex and the expectations exist, then

$$\varphi(\mathbb{E}X) \leq \mathbb{E}[\varphi(X)].$$

Geometrically, a convex graph lies above every supporting line. Applying this pointwise and taking expectations proves the result.

Important consequences include

$$(\mathbb{E}X)^2 \leq \mathbb{E}X^2, \text{ and } \mathbb{E}(1/X) \geq 1/\mathbb{E}X \quad (X > 0),$$

and the concavity reversal $\mathbb{E} \log X \leq \log \mathbb{E}X$.

12.4 Chernoff's method

For any $t > 0$, Markov applied to e^{tX} gives

$$\mathbb{P}(X \geq a) \leq e^{-ta} \mathbb{E}e^{tX}.$$

Optimizing over t yields the Chernoff bound. If $S_n \sim \text{Bin}(n, p)$ and $a > np$,

$$\mathbb{P}(S_n \geq a) \leq \inf_{t>0} e^{-ta} (1 - p + pe^t)^n.$$

Writing $a = nq$ produces the information-theoretic form

$$\mathbb{P}(S_n \geq nq) \leq e^{-nD(q\|p)},$$

where $D(q\|p) = q \log(q/p) + (1 - q) \log((1 - q)/(1 - p))$.

If independent $X_i \in [a_i, b_i]$ almost surely, then

$$\mathbb{P}\left(\sum_i (X_i - \mathbb{E}X_i) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_i (b_i - a_i)^2}\right).$$

The proof combines the exponential Markov step with a bound on the moment-generating function of a bounded centered variable.

12.5 The probabilistic method

To prove an object exists, choose an object randomly and show that the probability of being good is positive, or that the expected number of defects is less than one.

Order the vertices of a graph uniformly at random and select a vertex if it appears before all its neighbors. Selected vertices form an independent set. A vertex v is selected with

probability $1/(d(v) + 1)$, so some independent set has size at least

$$\sum_v \frac{1}{d(v) + 1} \geq \frac{n}{\bar{d} + 1},$$

where the last step is Jensen's inequality.

Use Markov when you know only a nonnegative mean; Chebyshev when you know a variance; Chernoff/Hoeffding when you have independence and exponential moments or boundedness; a union bound when failure is a union of manageable events.

Chapter check

Prove Markov and Chebyshev in three lines. Derive the exponential-Markov template. Explain why a small expected number of defects proves existence but does not necessarily give an efficient construction.

CHAPTER 13

Generating and Characteristic Functions

Learning objectives

Encode a distribution by a transform, extract moments, turn independent sums into products, and understand why characteristic functions are the natural tool for convergence in distribution.

13.1 Probability-generating functions

For a nonnegative integer-valued X , define

$$G_X(s) = \mathbb{E}[s^X] = \sum_{k=0}^{\infty} \mathbb{P}(X = k) s^k, \quad |s| \leq 1.$$

The coefficients recover the mass function, and when derivatives exist,

$$G'_X(1) = \mathbb{E}X, \quad G''_X(1) = \mathbb{E}[X(X-1)].$$

Thus $\text{Var}(X) = G''_X(1) + G'_X(1) - G'_X(1)^2$.

If X, Y are independent and nonnegative integer-valued, then

$$G_{X+Y}(s) = G_X(s)G_Y(s).$$

Proof. Independence factorizes $\mathbb{E}[s^{X+Y}] = \mathbb{E}[s^X s^Y]$. □

Key examples are

$$G_{\text{Bin}(n,p)}(s) = (1-p+ps)^n, \quad G_{\text{Pois}(\lambda)}(s) = e^{\lambda(s-1)}.$$

The product form immediately proves binomial and Poisson addition.

13.2 Random sums and branching

If $S_N = X_1 + \dots + X_N$, where N is independent of iid nonnegative integer-valued X_i , then

$$G_{S_N}(s) = G_N(G_X(s)).$$

Composition, rather than multiplication, is the signature of a random number of summands.

Let each individual independently produce X children with PGF G . If q is the extinction probability starting from one ancestor, conditioning on the first generation gives

$$q = G(q).$$

The extinction probability is the smallest solution in $[0, 1]$. Convexity of G shows $q = 1$ when $\mathbb{E}X \leq 1$, while $q < 1$ is possible when $\mathbb{E}X > 1$.

13.3 Moment-generating functions

The moment-generating function is

$$M_X(t) = \mathbb{E}e^{tX}.$$

If finite on an open interval containing zero, it determines the distribution and satisfies $M_X^{(k)}(0) = \mathbb{E}X^k$. It may fail to exist away from zero: the lognormal distribution has all positive moments but no finite MGF for $t > 0$.

For a normal variable,

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right).$$

This formula plus multiplication proves stability of normal sums.

13.4 Characteristic functions

The characteristic function

$$\varphi_X(t) = \mathbb{E}e^{itX}$$

exists for every real-valued random variable because $|e^{itX}| = 1$. It uniquely determines the distribution. Independence gives $\varphi_{X+Y} = \varphi_X \varphi_Y$.

If $\varphi_{X_n}(t) \rightarrow \varphi(t)$ pointwise and φ is continuous at zero and is a characteristic function, then X_n converges in distribution to the variable with characteristic function φ . The converse also holds.

This theorem turns convergence of distributions into ordinary pointwise convergence of functions and is a standard route to the central limit theorem.

If $X_n \sim \text{Bin}(n, \lambda/n)$, then

$$G_{X_n}(s) = \left(1 + \frac{\lambda(s-1)}{n}\right)^n \rightarrow e^{\lambda(s-1)},$$

the PGF of $\text{Pois}(\lambda)$. Hence X_n converges in distribution to $\text{Pois}(\lambda)$.

Chapter check

Compute the PGFs of geometric and negative-binomial variables. Explain why multiplication encodes fixed sums while composition encodes random sums. State one advantage of characteristic functions over MGFs.

CHAPTER 14

Convergence and Limit Theorems

Learning objectives

Distinguish four modes of convergence, prove a weak law with Chebyshev, standardize sums for the central limit theorem, and choose between exact, Poisson, and normal approximations.

14.1 Four modes of convergence

Let X_n, X be random variables.

- $X_n \rightarrow X$ almost surely if $\mathbb{P}(\lim_n X_n = X) = 1$.
- $X_n \rightarrow X$ in probability if $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.
- $X_n \rightarrow X$ in L^p if $\mathbb{E}|X_n - X|^p \rightarrow 0$.
- $X_n \Rightarrow X$ in distribution if $F_{X_n}(x) \rightarrow F_X(x)$ at every continuity point of F_X .

The standard implications are

$$X_n \rightarrow X \text{ a.s.} \implies X_n \rightarrow X \text{ in probability} \implies X_n \Rightarrow X,$$

and L^p convergence implies convergence in probability by Markov. None of the reverse implications holds without extra assumptions.

Let $X_n = n$ with probability $1/n$ and zero otherwise. Then $X_n \rightarrow 0$ in probability because $\mathbb{P}(|X_n| > \varepsilon) = 1/n$ for large n , but $\mathbb{E}X_n = 1$. Rare large values prevent convergence in mean; convergence in probability alone does not control tails.

14.2 The weak law of large numbers

If $X_1, X_2, \dots$ are iid with mean μ and finite variance σ^2 , then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \rightarrow \mu \quad \text{in probability.}$$

Proof. Independence gives $\mathbb{E}\bar{X}_n = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev yields

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2} \rightarrow 0.$$

□

The law of large numbers identifies a deterministic limit; it does not state the shape of the random error.

14.3 The central limit theorem

If X_i are iid with mean μ and variance $0 < \sigma^2 < \infty$, then

$$\frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \Rightarrow \mathcal{N}(0, 1).$$

Equivalently, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \Rightarrow \mathcal{N}(0, 1)$.

The CLT says typical fluctuations of the sample mean are of order $n^{-1/2}$ and become approximately normal. The theorem concerns standardized errors; the unscaled mean converges to a constant by the LLN.

14.3.1 Characteristic-function proof sketch

For centered variance-one variables, Taylor expansion near zero gives

$$\varphi_X(t) = 1 - \frac{t^2}{2} + o(t^2).$$

The characteristic function of $S_n/\sqrt{n}$ is

$$\varphi_X(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(n^{-1})\right)^n \rightarrow e^{-t^2/2},$$

the characteristic function of $\mathcal{N}(0, 1)$. Lévy's theorem completes the argument.

14.4 Approximations

For $S \sim \text{Bin}(n, p)$:

- use $\text{Pois}(np)$ when n is large, p is small, and np is moderate;
- use $\mathcal{N}(np, np(1-p))$ when both np and $n(1-p)$ are comfortably large;
- use a continuity correction, e.g.

$$\mathbb{P}(S \leq k) \approx \Phi\left(\frac{k + 1/2 - np}{\sqrt{np(1-p)}}\right).$$

14.5 Continuous mapping and Slutsky

If $X_n \rightarrow X$ in probability and g is continuous, then $g(X_n) \rightarrow g(X)$ in probability. If $X_n \Rightarrow X$ and $Y_n \rightarrow c$ in probability, then

$$X_n + Y_n \Rightarrow X + c, \text{ and } X_n Y_n \Rightarrow cX.$$

These results justify replacing unknown parameters in asymptotic standard errors by consistent estimates.

Write the exact mean and variance of the sum, subtract the mean, divide by the standard deviation, and only then use Φ . Many errors come from standardizing by n instead of $\sqrt{n}$.

Chapter check

Give counterexamples to two reverse implications among the convergence modes. Prove the weak law. Explain the continuity correction geometrically.

CHAPTER 15

Random Walks and Markov Chains

Learning objectives

Set up hitting-probability and hitting-time recurrences, compute multistep transition probabilities, identify stationary distributions, and understand the conditions behind long-run convergence.

15.1 Random walks

Let $X_i \in \{-1, +1\}$ be independent with $\mathbb{P}(X_i = 1) = p$ and define $S_n = S_0 + \sum_{i=1}^n X_i$. Then

$$\mathbb{E}S_n = S_0 + n(2p - 1), \quad \text{Var}(S_n) = 4np(1 - p).$$

Even for a fair walk, the typical displacement is $O(\sqrt{n})$ although the maximum possible displacement is n .

15.2 Gambler's ruin

The walk moves on $\{0, 1, \dots, N\}$ and stops at 0 or N . Let h_k be the probability of reaching N before 0 from k . First-step analysis gives

$$h_k = ph_{k+1} + qh_{k-1}, \quad h_0 = 0, \quad h_N = 1.$$

Solving the second-order recurrence yields

$$h_k = \begin{cases} k/N, & p = q = 1/2, \\ \frac{1 - (q/p)^k}{1 - (q/p)^N}, & p \neq q. \end{cases}$$

For the fair walk, the expected absorption time t_k satisfies

$$t_k = 1 + \frac{1}{2}t_{k-1} + \frac{1}{2}t_{k+1}, \quad t_0 = t_N = 0,$$

whose solution is $t_k = k(N - k)$.

15.3 The Markov property

Definition 15.1: Markov chain

A discrete-time process (X_n) on a state space S is a Markov chain if

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i).$$

For a time-homogeneous chain, this probability is p_{ij} , independent of n .

The matrix $P = (p_{ij})$ is row-stochastic. If the initial row distribution is μ_0 , then

$$\mu_n = \mu_0 P^n.$$

The Chapman–Kolmogorov equation is $P^{m+n} = P^m P^n$.

15.4 Communication and periodicity

State i reaches j if $(P^n)_{ij} > 0$ for some n . States communicate if each reaches the other. A chain is irreducible if all states communicate. The period of i is the greatest common divisor of return times n with $(P^n)_{ii} > 0$. In an irreducible chain all states have the same period.

15.5 Stationary distributions

Definition 15.2: Stationarity

A distribution π is stationary if

$$\pi P = \pi.$$

Every finite irreducible chain has a unique stationary distribution. If it is also aperiodic, then for every initial distribution μ_0 ,

$$\mu_0 P^n \rightarrow \pi.$$

Stationarity and convergence are different claims. A two-state chain that alternates deterministically has stationary distribution $(1/2, 1/2)$ but does not converge from a point mass because it has period two.

Let

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \quad a, b > 0.$$

Solving $\pi P = \pi$ and $\pi_1 + \pi_2 = 1$ gives

$$\pi = \left(\frac{b}{a+b}, \frac{a}{a+b} \right).$$

The balance equation $\pi_1 a = \pi_2 b$ says long-run flow across the two-state boundary is equal in both directions.

Chapter check

Derive gambler's ruin by solving the recurrence. Give a chain with a stationary law but no convergence from a fixed state. Explain the roles of irreducibility and aperiodicity.

Part IV

Mathematical Statistics

CHAPTER 16

Samples, Models, and Statistics

Learning objectives

Formulate a statistical experiment, distinguish parameters from statistics, analyze empirical summaries, and understand identifiability, sampling distributions, and the empirical CDF.

16.1 From probability to inference

Probability starts with a known distribution and predicts data. Statistics starts with data and reasons about an unknown distribution or parameter. A parametric model is a family

$$\mathcal{P} = \{P_\theta : \theta \in \Theta\}.$$

An iid sample $X_1, \dots, X_n$ means the joint law under θ is $P_\theta^{\otimes n}$.

Definition 16.1: Statistic and estimator

A statistic is a measurable function $T(X_1, \dots, X_n)$ whose formula does not involve the unknown parameter. An estimator is a statistic used to estimate a target $g(\theta)$.

An expression containing the symbol θ may still be a random variable but is not a usable statistic unless θ is known or eliminated.

16.2 Identifiability

Definition 16.2: Identifiable model

A model is identifiable if $P_{\theta_1} = P_{\theta_2}$ implies $\theta_1 = \theta_2$.

No amount of data can distinguish parameter values that define the same distribution. For example, parameterizing $\mathcal{N}(0, \theta^2)$ by $\theta \in \mathbb{R}$ is not identifiable because θ and $-\theta$ give the same law.

16.3 Sample mean and variance

For iid observations with mean μ and variance σ^2 ,

$$\bar{X} = \frac{1}{n} \sum_i X_i, \quad S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2.$$

Then $\mathbb{E}\bar{X} = \mu$, $\text{Var}(\bar{X}) = \sigma^2/n$, and $\mathbb{E}S^2 = \sigma^2$.

For every constant a ,

$$\sum_{i=1}^n (X_i - a)^2 = \sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - a)^2.$$

Proof. Write $X_i - a = (X_i - \bar{X}) + (\bar{X} - a)$ and expand. The cross term vanishes because $\sum_i (X_i - \bar{X}) = 0$. $\square$

Taking $a = \mu$ and expectations proves the unbiasedness of S^2 .

16.4 Order statistics and empirical distributions

Let $X_{(1)} \leq \dots \leq X_{(n)}$ be the ordered sample. The empirical CDF is

$$F_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{\{X_i \leq x\}}.$$

For each fixed x , $nF_n(x) \sim \text{Bin}(n, F(x))$, so

$$\mathbb{E}F_n(x) = F(x), \quad \text{Var}(F_n(x)) = \frac{F(x)(1 - F(x))}{n}.$$

The Glivenko–Cantelli theorem strengthens pointwise convergence to

$$\sup_x |F_n(x) - F(x)| \rightarrow 0 \quad \text{almost surely.}$$

16.5 Sampling distributions

The distribution of a statistic under repeated samples is its sampling distribution. It quantifies uncertainty. For a normal sample,

$$\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n),$$

and, more remarkably,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2, \quad \bar{X} \text{ is independent of } S^2.$$

These exact facts power classical confidence intervals and tests.

Before data are observed, $\bar{X}$ is random. After observing $x_1, \dots, x_n$, the value $\bar{x}$ is fixed. Confidence statements concern the random procedure across hypothetical repetitions.

Chapter check

Explain identifiability. Prove the ANOVA identity. Derive the mean and variance of $F_n(x)$ for fixed x .

CHAPTER 17

Point Estimation

Learning objectives

Evaluate estimators by bias, variance, MSE, consistency, and asymptotic distribution; construct estimators by moments and maximum likelihood; and check boundaries and invariance carefully.

17.1 Risk and quality criteria

For an estimator T of $g(\theta)$,

$$\text{Bias}_\theta(T) = \mathbb{E}_\theta T - g(\theta).$$

Under squared-error loss, the risk is the mean squared error

$$\text{MSE}_\theta(T) = \mathbb{E}_\theta[(T - g(\theta))^2] = \text{Var}_\theta(T) + \text{Bias}_\theta(T)^2.$$

Unbiasedness is attractive but not decisive: a slightly biased estimator can have much smaller MSE.

An estimator T_n is consistent if $T_n \rightarrow g(\theta)$ in probability for every θ . Consistency concerns large samples; it does not rank estimators at a fixed n .

17.2 The method of moments

If $\theta = (\theta_1, \dots, \theta_k)$ and the first k population moments determine θ , set

$$\frac{1}{n} \sum_{i=1}^n X_i^r = \mathbb{E}_\theta[X^r], \quad r = 1, \dots, k,$$

and solve. The law of large numbers often makes the resulting estimator consistent.

For a gamma distribution with shape α and rate λ ,

$$\mu = \alpha/\lambda, \quad \sigma^2 = \alpha/\lambda^2.$$

Replacing these by $\bar{X}$ and the empirical second central moment M_2 gives

$$\hat{\alpha}_{\text{MM}} = \frac{\bar{X}^2}{M_2}, \quad \hat{\lambda}_{\text{MM}} = \frac{\bar{X}}{M_2}.$$

17.3 Maximum likelihood

Given observed data $x = (x_1, \dots, x_n)$, the likelihood is

$$L(\theta; x) = \prod_{i=1}^n f_{\theta}(x_i),$$

viewed as a function of θ . The maximum-likelihood estimator (MLE) maximizes L , equivalently the log-likelihood

$$\ell(\theta; x) = \sum_i \log f_{\theta}(x_i).$$

For $X_i \stackrel{\text{iid}}{\sim} \text{Bern}(p)$ and $S = \sum_i X_i$,

$$\ell(p) = S \log p + (n - S) \log(1 - p).$$

The derivative vanishes at $\hat{p} = S/n = \bar{X}$, and concavity shows this is the maximum, with boundary interpretations when $S = 0$ or $S = n$.

For $X_i \stackrel{\text{iid}}{\sim} \text{Unif}[0, \theta]$,

$$L(\theta) = \theta^{-n} \mathbf{1}_{\{\theta \geq X_{(n)}\}}.$$

On the feasible region likelihood decreases with θ , so

$$\hat{\theta}_{\text{MLE}} = X_{(n)}.$$

Differentiating $-n \log \theta$ and ignoring the support would produce no solution. Boundaries are part of the model.

17.4 Invariance and transformation

If $\hat{\theta}$ is an MLE for θ , then $g(\hat{\theta})$ is an MLE for $g(\theta)$, with the appropriate profile interpretation when g is not one-to-one. This invariance is useful but does not generally preserve unbiasedness.

17.5 Asymptotic behavior

Under regularity and identifiability conditions, MLEs are typically consistent and

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, I(\theta)^{-1}),$$

where $I(\theta)$ is Fisher information per observation. The formula anticipates both uncertainty quantification and asymptotic efficiency.

Common pitfall. Solving the score equation finds stationary points, not necessarily the global maximum. Check concavity, boundaries, parameter constraints, and whether the likelihood is unbounded.

Chapter check

Derive the MLE for the exponential rate. Show that the sample mean is consistent for the population mean. Explain why unbiasedness and consistency are logically independent.

CHAPTER 18

Sufficiency, Completeness, and Fisher Information

Learning objectives

Compress samples without losing parameter information, apply the factorization criterion, use Rao–Blackwell improvement, compute Fisher information, and recognize when Cramér–Rao regularity fails.

18.1 Sufficient statistics

Definition 18.1: Sufficiency

A statistic $T(X)$ is sufficient for θ if the conditional distribution of the full sample X given T does not depend on θ .

Sufficiency is an information statement: once T is known, the remainder of the data carries no additional information about the parameter within the assumed model.

For a dominated model, T is sufficient if and only if the joint density or mass function can be written

$$f_{\theta}(x) = g_{\theta}(T(x))h(x),$$

where h does not depend on θ .

For Bernoulli data,

$$f_p(x_1, \dots, x_n) = p^{\sum x_i} (1-p)^{n-\sum x_i},$$

so $T = \sum_i X_i$ is sufficient. The order of successes contains no additional information about p .

18.2 Rao–Blackwell improvement

Let U estimate $g(\theta)$ with finite second moment, and let T be sufficient. Then

$$U^* = \mathbb{E}[U | T]$$

has the same expectation as U and no larger variance. Under squared error it has no larger risk.

Proof. The tower property preserves the mean, and total variance gives

$$\text{Var}(U) = \mathbb{E} \text{Var}(U | T) + \text{Var}(\mathbb{E}[U | T]) \geq \text{Var}(U^*).$$

□

18.3 Completeness

A statistic T is complete if $\mathbb{E}_\theta h(T) = 0$ for every θ implies $h(T) = 0$ almost surely. If a complete sufficient statistic exists, an unbiased function of it is the unique unbiased estimator of its target with minimum variance. This is the Lehmann–Scheffé principle.

For example, the sum of iid Poisson observations is complete and sufficient for λ . Completeness often arises in full exponential families.

18.4 Score and Fisher information

The score for one observation is

$$U_\theta(X) = \frac{\partial}{\partial \theta} \log f_\theta(X).$$

Under regularity conditions, $\mathbb{E}_\theta U_\theta = 0$. Fisher information is

$$I(\theta) = \mathbb{E}_\theta [U_\theta(X)^2] = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X) \right].$$

For independent observations, information adds: $I_n(\theta) = nI(\theta)$.

For $X \sim \text{Bern}(p)$,

$$U_p(X) = \frac{X}{p} - \frac{1-X}{1-p} = \frac{X-p}{p(1-p)}.$$

Therefore

$$I(p) = \frac{\text{Var}(X)}{p^2(1-p)^2} = \frac{1}{p(1-p)}.$$

The sample proportion has variance $p(1-p)/n = 1/[nI(p)]$.

18.5 The Cramér–Rao lower bound

Under regularity conditions, every unbiased estimator T of $g(\theta)$ satisfies

$$\text{Var}_\theta(T) \geq \frac{[g'(\theta)]^2}{nI(\theta)}.$$

The proof applies Cauchy–Schwarz to T and the score after differentiating $\mathbb{E}_\theta T = g(\theta)$ under the integral sign.

For $\text{Unif}[0, \theta]$, differentiation under the integral misses a moving boundary; the score need not have mean zero. Do not apply Cramér–Rao mechanically when the support depends on θ .

Chapter check

Factor the likelihoods for Bernoulli, Poisson, and normal samples. Prove Rao–Blackwell from total variance. Compute information for an exponential rate.

CHAPTER 19

Confidence Intervals

Learning objectives

Interpret coverage correctly, construct intervals from pivots and asymptotic normality, distinguish known and unknown variance, and connect interval width to sample size.

19.1 Coverage, not posterior probability

Definition 19.1: Confidence interval

A random interval $C(X) = [L(X), U(X)]$ has confidence level $1 - \alpha$ for θ if

$$\mathbb{P}_\theta(\theta \in C(X)) \geq 1 - \alpha \quad \text{for every } \theta \in \Theta.$$

In a frequentist model, θ is fixed and the interval is random before sampling. After data are observed, the reported interval either contains θ or does not; 95% describes the long-run procedure, not a posterior probability for this one fixed interval.

19.2 Pivotal quantities

A pivot is a function of data and parameter whose distribution does not depend on unknown parameters. If

$$\mathbb{P}(a \leq Q(X, \theta) \leq b) = 1 - \alpha,$$

solve the inequalities for θ to obtain a confidence interval.

For $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ ,

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1).$$

Thus

$$\mu \in \left[\bar{X} - z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$$

with probability $1 - \alpha$.

19.3 Unknown variance and Student's law

For a normal sample with unknown σ ,

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Therefore the exact interval is

$$\bar{X} \pm t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}}.$$

Replacing z by t accounts for estimating the standard deviation. As n grows, t quantiles approach normal quantiles.

19.4 Intervals for a proportion

For Bernoulli data, $\hat{p} = \bar{X}$ is approximately normal with standard error $\sqrt{p(1-p)/n}$. The Wald interval substitutes $\hat{p}$:

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

It performs poorly for small samples or proportions near zero or one and may leave $[0, 1]$. Score-based Wilson intervals or exact binomial intervals are preferable in those regimes.

19.5 Asymptotic intervals

If

$$\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, V(\theta))$$

and $\hat{V}$ is consistent, then an approximate interval is

$$\hat{\theta} \pm z_{1-\alpha/2} \sqrt{\hat{V}/n}.$$

Slutsky's theorem justifies replacing the unknown variance by a consistent estimate.

19.6 Planning sample size

For a normal mean with known σ , desired half-width m requires

$$n \geq \left(\frac{z_{1-\alpha/2} \sigma}{m} \right)^2.$$

Halving the margin of error requires approximately four times the sample size. Increasing confidence also widens the interval.

An interval without its confidence level, target parameter, and assumptions is incomplete. In applications, also report the point estimate and sample size.

Chapter check

Derive the known-variance normal interval by inverting a pivot. Explain why a 95% interval is not a 95% posterior probability statement. Derive the $1/\sqrt{n}$ width law.

CHAPTER 20

Hypothesis Testing and Likelihood Ratios

Learning objectives

Formulate null and alternative hypotheses, control type-I error, interpret power and p -values, apply Neyman–Pearson, and connect tests with confidence sets.

20.1 Tests as decision rules

A test is a function $\varphi(X) \in [0, 1]$, interpreted as the probability of rejecting H_0 after observing X . Nonrandomized tests take values zero or one. For hypotheses

$$H_0 : \theta \in \Theta_0, \quad H_1 : \theta \in \Theta_1,$$

the size is

$$\sup_{\theta \in \Theta_0} \mathbb{E}_\theta \varphi(X).$$

A level- α test has size at most α . Its power function is $\beta(\theta) = \mathbb{E}_\theta \varphi(X)$ for alternatives.

Type-I error is rejecting a true null; type-II error is failing to reject a false null. Reducing both uniformly is usually impossible at fixed sample size.

20.2 The Neyman–Pearson lemma

For testing simple $H_0 : f = f_0$ against simple $H_1 : f = f_1$, a most powerful level- α test rejects for large values of

$$\Lambda(X) = \frac{f_1(X)}{f_0(X)},$$

with possible randomization on the boundary to achieve exact size.

The likelihood ratio ranks data by how much more compatible they are with H_1 than with H_0 . The lemma is a precise statement that this ranking is optimal.

Let $X \sim \mathcal{N}(\theta, 1)$ and test $H_0 : \theta = 0$ against $H_1 : \theta = \delta > 0$. The log-likelihood ratio is

$$\delta X - \frac{\delta^2}{2},$$

which increases in X . The most powerful level- α test rejects when $X > z_{1-\alpha}$. Its power at $\theta = \delta$ is

$$1 - \Phi(z_{1-\alpha} - \delta).$$

20.3 Composite alternatives and generalized likelihood ratios

For composite hypotheses, a common statistic is

$$\lambda(X) = \frac{\sup_{\theta \in \Theta_0} L(\theta; X)}{\sup_{\theta \in \Theta} L(\theta; X)}.$$

Small values contradict the null. Under regularity and with r constraints, Wilks' theorem gives

$$-2 \log \lambda(X) \Rightarrow \chi_r^2$$

under H_0 .

20.4 p -values

A valid p -value is a statistic $p(X)$ satisfying

$$\mathbb{P}_\theta(p(X) \leq u) \leq u \quad \text{for all } \theta \in \Theta_0, 0 \leq u \leq 1.$$

Operationally, it is the smallest significance level at which the observed result would be rejected by the chosen nested family of tests.

It is not $\mathbb{P}(H_0 | X)$, not the probability the result occurred “by chance,” and not a measure of effect size. A tiny effect can be highly significant in a huge sample.

20.5 Duality with confidence sets

For each θ_0 , let $A(\theta_0)$ be the acceptance region of a level- α test of $H_0 : \theta = \theta_0$. Then

$$C(X) = \{\theta_0 : X \in A(\theta_0)\}$$

is a $(1 - \alpha)$ confidence set. Conversely, a confidence set induces tests by rejecting values outside it.

20.6 Multiple testing

Testing m hypotheses each at level α can greatly exceed family-wise error α . Bonferroni tests each at level α/m ; by the union bound, the probability of at least one false rejection is at most α . False-discovery-rate procedures answer a different question and may be more powerful in large-scale screening.

One-sided and two-sided tests answer different questions. Selecting the direction after inspecting the estimate invalidates the nominal level unless the selection is accounted for.

Chapter check

Define size, power, and p -value precisely. Derive the normal one-sided test from Neyman–Pearson. Explain test–interval duality.

Classical Parametric and Goodness-of-Fit Tests

Learning objectives

Choose among z , t , χ^2 , and F procedures, compare one and two samples, perform a chi-square goodness-of-fit test, and audit assumptions before reporting a conclusion.

21.1 One-sample location tests

For $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with known σ , testing $H_0 : \mu = \mu_0$ uses

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0, 1) \quad (H_0).$$

With unknown σ , replace it by S :

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}.$$

The exact t law relies on normal sampling. For large samples, an asymptotic t approximation can be robust, but severe skewness and heavy tails still matter.

21.2 Two samples and pairing

For independent samples, the difference $\bar{X} - \bar{Y}$ estimates $\mu_X - \mu_Y$. Welch's statistic is

$$T = \frac{\bar{X} - \bar{Y} - \Delta_0}{\sqrt{S_X^2/n + S_Y^2/m}},$$

with an approximate Welch–Satterthwaite number of degrees of freedom. It does not assume equal variances.

Paired data require a different model. If observations are matched pairs (X_i, Y_i) , analyze differences $D_i = X_i - Y_i$ with a one-sample procedure. Treating paired measurements as independent discards covariance and can severely reduce power.

21.3 Variance procedures

For a normal sample,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

This yields exact intervals and tests for σ^2 . For independent normal samples,

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{n-1, m-1}.$$

Variance tests are notably sensitive to nonnormality.

21.4 Pearson chi-square tests

Suppose category counts are $O_1, \dots, O_k$ and null expected counts are $E_i = np_i$. Pearson's statistic is

$$X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}.$$

Under standard conditions it is approximately χ_{k-1}^2 when the p_i are fully specified. If r parameters are estimated from the same counts, the degrees of freedom typically become $k - 1 - r$.

A die is rolled n times. Under fairness $E_i = n/6$. Compute

$$X^2 = \sum_{i=1}^6 \frac{(O_i - n/6)^2}{n/6}$$

and compare to χ_5^2 . The test detects departures in any categorical direction but does not say which alternative is scientifically important; residuals $(O_i - E_i)/\sqrt{E_i}$ help diagnose the pattern.

21.5 Independence in contingency tables

For an $r \times c$ table, the null of independence predicts

$$E_{ij} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{n}.$$

Pearson's statistic is asymptotically $\chi_{(r-1)(c-1)}^2$. Small expected cell counts call for category pooling or exact methods.

21.6 Assumption audit

Before using a classical test, check:

- independence or the actual dependence structure;
- whether pairing, clustering, or repeated measurements are present;
- normality when an exact small-sample claim depends on it;
- whether asymptotic expected counts are sufficiently large;
- whether the test direction and analysis plan were specified in advance.

Chapter check

Explain the difference between paired and independent samples. State the exact null laws of the normal-theory t , χ^2 , and F statistics. Derive expected counts under table independence.

CHAPTER 22

Linear Regression and Correlation

Learning objectives

Derive least squares geometrically and algebraically, interpret coefficients conditionally, quantify uncertainty, diagnose residuals, and separate prediction, association, and causation.

22.1 The linear model

In matrix form,

$$y = X\beta + \varepsilon, \quad \mathbb{E}(\varepsilon \mid X) = 0, \quad \text{Cov}(\varepsilon \mid X) = \sigma^2 I.$$

The design matrix X contains predictors and usually a column of ones. Least squares minimizes

$$\|y - Xb\|_2^2.$$

If X has full column rank, the unique minimizer is

$$\hat{\beta} = (X^\top X)^{-1} X^\top y.$$

The fitted vector $\hat{y} = X\hat{\beta}$ is the orthogonal projection of y onto the column space of X , and the residual $e = y - \hat{y}$ satisfies $X^\top e = 0$.

Proof. Differentiate the quadratic objective to obtain the normal equations $X^\top Xb = X^\top y$. Full rank makes $X^\top X$ invertible. The normal equations are precisely the orthogonality condition. $\square$

22.2 Simple regression

For $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$,

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

With an intercept, residuals sum to zero and the fitted line passes through $(\bar{x}, \bar{y})$.

22.3 Sampling properties

Conditional on X ,

$$\mathbb{E}\hat{\beta} = \beta, \quad \text{Cov}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}.$$

Under normal errors, standardized coefficient estimates have exact t distributions after replacing σ by the residual standard error. Without normality, large-sample robust standard errors are often used.

22.4 Goodness of fit

With an intercept,

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i (y_i - \hat{y}_i)^2.$$

$\sum_i \{y_i - \bar{y}\}^2$
TSS

$\sum_i \{\hat{y}_i - \bar{y}\}^2$
ESS

$\sum_i \{y_i - \hat{y}_i\}^2$
RSS

The coefficient of determination is $R^2 = 1 - \text{RSS}/\text{TSS}$. Adding predictors cannot increase training RSS, so in-sample R^2 never decreases even for useless predictors. Predictive evaluation therefore requires held-out or cross-validated data.

22.5 Interpretation and omitted variables

In multiple regression, β_j is the change in conditional mean associated with a one-unit increase in predictor j while holding the other included predictors fixed. This is not automatically a causal effect. If an omitted variable influences both predictor and response, the zero-conditional-mean assumption can fail.

Within each of two departments, admission rates may be higher for group A than group B, yet the aggregate rate for A may be lower if A applies disproportionately to the more selective department. Regression adjustment and stratification make the hidden weighting visible; neither by itself guarantees a causal interpretation.

22.6 Diagnostics

Inspect residuals against fitted values and predictors. Curvature suggests nonlinearity; a funnel shape suggests heteroskedasticity; isolated large residuals suggest outliers; high leverage makes a point capable of changing the fit substantially. Time order, clusters, and spatial structure must be checked for dependence.

A model can predict well using noncausal proxies and can estimate a causal effect poorly despite high R^2 . State the goal before choosing the validation and interpretation.

Chapter check

Derive the normal equations. Explain projection geometry. Prove that adding a predictor cannot increase training RSS. State why high correlation does not establish causation.

CHAPTER 23

Nonparametric and Asymptotic Statistics

Learning objectives

Use the empirical distribution as a model-free estimator, understand rank and permutation tests, apply bootstrap logic, and use the delta method for smooth transformations.

23.1 Distribution-free thinking

Parametric methods gain efficiency by imposing structure. Nonparametric methods use weaker assumptions and are often more robust, but no method is assumption free. Exchangeability, continuity, symmetry, or smoothness typically replaces a named parametric family.

23.2 The empirical CDF and Kolmogorov distance

The empirical CDF F_n estimates the entire distribution. The Dvoretzky–Kiefer–Wolfowitz inequality states

$$\mathbb{P} \left(\sup_x |F_n(x) - F(x)| > \varepsilon \right) \leq 2e^{-2n\varepsilon^2}.$$

It gives a finite-sample uniform confidence band and immediately implies the Glivenko–Cantelli theorem.

The Kolmogorov–Smirnov statistic

$$D_n = \sup_x |F_n(x) - F_0(x)|$$

tests a fully specified continuous null distribution. Estimating parameters from the same sample changes the null distribution.

23.3 Ranks and permutation tests

Under a continuous null of identical distributions, pooled sample ranks are distribution free. Rank-sum tests compare locations without relying on a normal model, though their interpretation depends on whether distributions differ only by location.

A permutation test chooses a statistic T and compares the observed value with values obtained by permuting labels under an exchangeability null. Conditional on the observed pooled outcomes, the permutation distribution is exact if the labels are exchangeable.

In a randomized experiment with a fixed number assigned to treatment, the sharp null of no individual treatment effect implies outcomes would be unchanged by reassignment. Recompute the difference in means over all assignments (or a large random subset). The fraction at least as extreme as observed is an exact randomization p -value.

23.4 Bootstrap

The nonparametric bootstrap samples $X_1^*, \dots, X_n^*$ with replacement from the empirical distribution F_n . Repeating a statistic $T^* = T(X^*)$ approximates its sampling distribution under F . Bootstrap standard errors, percentile intervals, and bias corrections are powerful, but the method can fail for nonsmooth statistics, extreme tails, boundary parameters, or dependent data unless adapted.

23.5 The delta method

If

$$\sqrt{n}(T_n - \theta) \Rightarrow \mathcal{N}(0, V)$$

and g is differentiable at θ , then

$$\sqrt{n}(g(T_n) - g(\theta)) \Rightarrow \mathcal{N}(0, [g'(\theta)]^2 V).$$

Proof. Taylor expansion gives

$$g(T_n) - g(\theta) = g'(\theta)(T_n - \theta) + o_p(|T_n - \theta|).$$

Multiply by $\sqrt{n}$ and apply Slutsky's theorem. □

For a vector parameter, the asymptotic covariance transforms as $Dg(\theta)VDg(\theta)^\top$.

If $\hat{p}$ is a sample proportion and $g(p) = \log[p/(1-p)]$, then $g'(p) = 1/[p(1-p)]$. Since $\sqrt{n}(\hat{p} - p) \Rightarrow \mathcal{N}(0, p(1-p))$,

$$\sqrt{n}\{g(\hat{p}) - g(p)\} \Rightarrow \mathcal{N}\left(0, \frac{1}{p(1-p)}\right).$$

Chapter check

State the exchangeability assumption behind a permutation test. Describe the bootstrap algorithm in three lines. Prove the delta method from Taylor expansion.

Part V

Admissions Synthesis

Admissions Problem-Solving Playbook

Learning objectives

Convert a dense problem into a model, select a method from structural signals, produce a rigorous short solution, and manage time on a selective admissions examination.

24.1 The five-pass reading protocol

- Pass 1. Objects.** What is random: a sequence, subset, permutation, graph, point, or stopping time?
- Pass 2. Mechanism.** Uniform selection, independent trials, sampling without replacement, a mixture, or a process evolving by steps?
- Pass 3. Target.** A probability, expectation, distribution, bound, existence statement, or estimator?
- Pass 4. Structure.** Symmetry, complement, indicators, conditioning, recurrence, convexity, or transform?
- Pass 5. Audit.** What assumptions justify equal likelihood, independence, interchange of sums, or an approximation?

24.2 Signal-to-method dictionary

Signal in the problem	First method to test
At least one; no occurrence	Complement or union bound
Expected number of matches, pairs, cycles	Indicators and linearity
Hidden source, type, or parameter	Total probability / Bayes
Time until a pattern or boundary	State recursion / first-step analysis
Random number of objects	Conditional expectation / PGF composition
Minimum or maximum	CDF or survival function
Sum of independent variables	Convolution or transforms
Existence without construction	Random construction and expectation
Rare events, large n	Poisson or exponential bound
Empirical average, large n	LLN / CLT after standardization
Unknown parameter from data	Likelihood, moments, sufficiency
Decision under uncertainty	Hypotheses, level, power, test statistic

24.3 Proof patterns worth memorizing

Do not memorize isolated answers; internalize short proof engines.

24.3.1 Indicator expansion

Write a count as $N = \sum_i \mathbf{1}_{A_i}$, then $\mathbb{E}N = \sum_i \mathbb{P}(A_i)$. Use pairwise products for the variance.

24.3.2 Condition on the first step

For an expected time, write $t_i = 1 + \sum_j p_{ij}t_j$. For a hitting probability, omit the one: $h_i = \sum_j p_{ij}h_j$.

24.3.3 Condition on a latent variable

Find Z such that the law of the target given Z is elementary. Apply total probability, total expectation, or total variance.

24.3.4 Symmetry by a map

A symmetry argument is rigorous when it identifies a probability-preserving bijection or group action mapping one event to another.

24.3.5 Existence by averaging

Define a random candidate and a nonnegative defect count. If $\mathbb{E}N < 1$, some candidate has no defects; if $\mathbb{E}V = m$, some candidate has value at least m .

24.4 Writing a solution

A strong short solution normally has four paragraphs:

1. define the sample space or random variables;
2. state the central identity or recurrence;
3. compute or solve it;
4. check the range, boundary cases, and assumptions.

Avoid narrative algebra. Every displayed equation should advance the argument.

Problem. Find the expected number of cycles in a uniform random permutation of n elements. Insert elements in increasing order. When element k is inserted into a uniform permutation of $1, \dots, k-1$, there are k equally likely insertion positions in cycle notation; exactly one starts a new cycle. Let I_k indicate that k starts a new cycle. Then $\mathbb{P}(I_k = 1) = 1/k$ and

$$C_n = \sum_{k=1}^n I_k, \text{ and } \mathbb{E}C_n = \sum_{k=1}^n \frac{1}{k} = H_n.$$

The argument defines the coupling, the indicator, and the probability; no independence claim is needed.

24.5 Time strategy

On a multi-problem exam, spend the first ten minutes classifying every problem. Secure complete solutions to accessible problems before investing heavily in a challenge. If stuck, write down a correct partial reduction: a recurrence, exact finite sum, or useful bound earns more than unexplained experimentation.

Leave time to audit:

- probabilities lie in $[0, 1]$ and masses sum to one;
- expectations have plausible units and scale;
- asymptotics agree with small or extreme cases;
- denominators are nonzero and supports are correct;
- a claimed independence is actually justified.

Admissions problems reward controlled invention: recognize a standard structure, adapt it cleanly, and explain why the adaptation is valid. The problem book is organized to train precisely that transition.

Chapter check

Take any problem you solved recently and rewrite its solution in the four-paragraph format. Identify the structural signal that should have suggested the decisive method.

Notation

Symbol	Meaning
$\mathbb{P}(A)$	probability of event A
$\mathbb{E}X, \text{Var}(X)$	expectation and variance
$\text{Cov}(X, Y), \rho(X, Y)$	covariance and correlation
F_X, f_X, p_X	distribution function, density, and mass function
$\mathbf{1}_A$	indicator of event A
$X_n \rightarrow X$ a.s., in L^p , in probability, in dis- tribution	standard modes of convergence
$\ell(\theta), I(\theta)$	log-likelihood and Fisher information
$\hat{\theta}, \text{Bias}(\hat{\theta})$	estimator and its bias

Suggested references

- D. Bertsekas and J. Tsitsiklis, *Introduction to Probability*.
- G. Grimmett and D. Stirzaker, *Probability and Random Processes*.
- S. Ross, *A First Course in Probability*.
- L. Wasserman, *All of Statistics*.
- G. Casella and R. Berger, *Statistical Inference*.
- A. Shiryaev, *Probability*, for a measure-theoretic continuation.